

1. Select all of the expressions that are equivalent to 10^{-2} .

☐ $-\left(\frac{1}{10}\right)^2$

☐ $\frac{1}{-20}$

☒ $\frac{1}{10^2}$

☐ -20

☐ $(-2)^{-10}$

☒ $\frac{1}{10} \cdot \frac{1}{10}$

☐ $-\left(\frac{1}{10} \cdot \frac{1}{10}\right)$

2. It is estimated that the novel *A Land Remembered* by Patrick Smith has 503,000 letters and the novel *War and Peace* by Leo Tolstoy has 3,200,000 letters.

- a. Write the number of letters in *A Land Remembered* using scientific notation.

Scientific Notation	5.03×10^5
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- b. Write the number of letters in *War and Peace* using scientific notation.

Scientific Notation	3.2×10^6
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3. In 2015, the value of the freight, in dollars, that came into Florida was 3.46×10^{11} . The value of the freight, in dollars, that flowed within Orlando was 1.73×10^{10} . How many times larger is the value of the freight, in dollars, that came into Florida than the freight that flowed within Orlando?

Comparison	20 times larger
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4. It is estimated that during the year 2018, all of bees in Florida flew a total distance of 1.39×10^{21} feet. Write the distance in standard form.

Standard Form	1,390,000,000,000,000,000,000 feet
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5. The speed of an average garden snail is estimated to be 8.056×10^{-6} miles per second. Write the speed in standard form.

Standard Form	0.000008056 miles per second
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6. The table shows the global plastic production, in milligrams (mg), for various years.

Year	Global Plastic Production (mg)
1953	1.058×10^5
1986	3.38×10^6
2015	1.352×10^7

- a. How many times smaller was the global plastic production in 1986 than 2015?

Comparison	4 times smaller
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- b. Determine the total amount of plastic, in milligrams, produced in 1953 and 1986.

Total	3.4858×10^6 mg
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- c. Determine the difference of the amount of plastic produced in 2015 and the amount of plastic produced in 1953.

Difference	1.34142×10^7 mg
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7. The Irukandji jellyfish and the Brazilian Wandering Spider are two of the most venomous animals in the world. A researcher recorded the size of an Irukandji jellyfish as 3.472×10^{-5} cubic feet. The researcher also recorded the size of a Brazilian Wandering Spider as 3.09×10^{-3} cubic feet. Determine the difference between the size of the spider and the size of the jellyfish. Write your answer using significant digits.

Difference	3.06×10^{-3} cubic feet
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8. An expression is shown.

$$(-6 \times 10^{-8})(1.9 \times 10^{-6})$$

Select all of the expressions that are equivalent to the given expression.

- [illegible]

9. An expression is shown.

$$(5.2 \times 10^{17}) - (9.8637 \times 10^{19})$$

Select all of the expressions that are equivalent to the given expression.

- ☐ -4.6637×10^2
- ☐ -985.85×10^{17}
- ☒ -9.8117×10^{19}
- ☐ -466.37
- ☐ $-98,585,000,000,000,000,000$
- ☒ $-98,117,000,000,000,000,000$

Determine the value of each expression. Show your work for partial credit. Express your answer using scientific notation and in standard form.

10.	$\frac{4.7 \times 10^{-4}}{-3.76 \times 10^{-6}}$	
	Scientific Notation	Standard Form
	-1.25×10^2	-125
11.	$(5 \times 10^{-5}) + (8 \times 10^{-6})$	
	Scientific Notation	Standard Form
	5.8×10^{-5}	0.000058
12.	$(2.65 \times 10^{11}) \div (5 \times 10^{13})$	
	Scientific Notation	Standard Form
	5.3×10^{-3}	0.0053

13.	$(2.9 \times 10^5)(6 \times 10^7)$	
	Scientific Notation	Standard Form
	1.74×10^{13}	17,400,000,000,000
14.	$(5 \times 10^{-1})(9 \times 10^{-3})$	
	Scientific Notation	Standard Form
	4.5×10^{-3}	0.0045